Touchscreens

They're all the rage, they're the buzzword, they're probably the hottest thing in tech right now!

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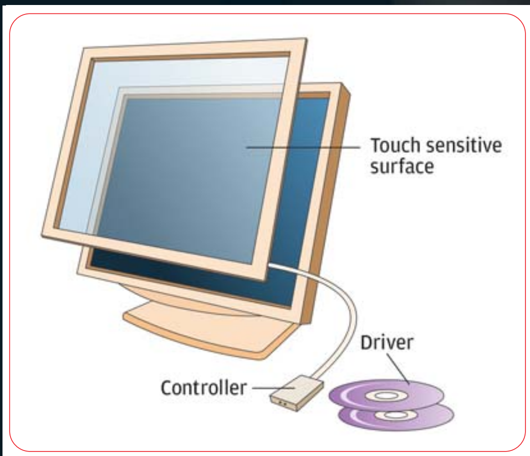
You've seen touchscreens at information kiosks, in museums, hotels, and ATMs, and you did marvel (at least the first time) the way they interacted with you. Touchscreen-based systems allow for easy navigation around a GUI-based environment. Tap on the screen twice for double clicks, and drag a finger on the screen to move the cursor. You can do almost everything the mouse does, and you do away with the additional device (the mouse)—leaving only the screen.

Falling prices and improved technology has fuelled the use of touchscreens in smaller consumer devices like mobiles, Tablet PCs, and handheld gaming consoles like the Nintendo DS.

As an aside, you don't need to buy a full-fledged touchscreen for touch-based interaction. Touchscreen components can be bought off the shelf (at least in developed countries) and used with an existing screen!

The Components

There are four popular touchscreen technologies we'll talk about, but all of them have three main components: the touch sensitive surface, the controller, and the software driver.



The three basic components of a touchscreen: the sensor, controller and driver

- The touch sensitive surface is an extremely durable and flexible glass or polymer touch-responsive surface, and this panel is placed over the viewable area of the screen. In most sensors, there is an electric current or signal going across the screen, and a touch on the surface causes a change in the signal depending on the touch sensor technology used. This change allows the controller to identify the location of the touch.
- The controller is an electronic device that acts as the intermediary between the screen and the computer. It interprets the electrical signals of the touch event to digital signals that a computer can understand. The controller can be integrated with the screen or housed externally.
- The software driver is an interpreter that converts what comes from the controller to information that the OS can understand. It is similar to a mouse driver, to take a common example.

Four Touchscreen Sensor Technologies

Resistive touchscreens

Based on an old technology, these are durable and resistant to humidity and liquid spills. But they offer limited clarity, and the surface can be easily damaged by sharp objects. Resistive touchscreens have two glass or acrylic layers. One of the layers is coated with thin conductive material, and the other is coated with resistive material. When in operation, electricity passes across the screen on the conductive layer; when pressure is applied, the two layers are pressed against each other and there is a change in the electric current due to the resistance brought in by the resistive layer. This change is registered and the location of touch is calculated.

Capacitive touchscreens

These are used for applications where clarity and precision is of concern—as in laptops and medical imaging. Such screens cannot be used with non-conductive inputs (like rubber-gloved hands or styli).

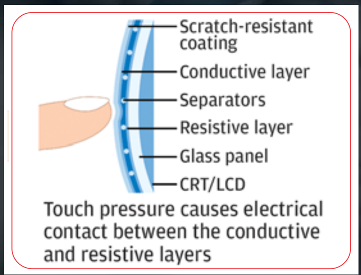
Here, we have a glass panel with a coat of charge-storing material on its surface. When the panel is touched, a small amount of charge is drawn at the point of contact. Circuits located at each corner of the screen measure the difference in charge and send the information to the controller for calculating the position of touch.

Surface Acoustic Wave Touchscreens

This newer technology uses ultrasonic waves that pass over the screen. When the panel is touched, there is a change in the frequency of the ultrasonic waves, and the receiver at the end of the panel registers this change. Since only glass is used with no coating, there is nothing that can wear out.

Other technologies

The above are the mainstream technologies; there are others as well. Strain gauge touch panels have springs mounted on the four corners; when touched, the change in the compression in



Resistive touchscreens—the most robust and affordable